Hull Tactical Market Prediction

Tactical Asset Allocation with Machine Learning

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Outline

The Kaggle Challenge

- **Goal**: Predict daily S&P 500 excess returns and convert to weights in $[0, 2]$
- **Metric**: Adjusted Sharpe ratio (penalizes risk and underperformance)
- **Data**: Anonymized macroeconomic, sentiment, and pricing features ($\sim$9k days)
- **Constraint**: Weights must be between 0 and 2; predictions evaluated daily

Why this matters

Tactical asset allocation adjusts market exposure based on predictive signals. The adjusted Sharpe metric is useful here because it penalizes both excess risk and falling behind the market.

Adjusted Sharpe Ratio

$$\text{adjusted\_sharpe} = \frac{\text{Sharpe}}{\text{vol\_penalty} \times \text{return\_penalty}}$$

$$\text{vol\_penalty} = 1 + \max\left(0,\ \frac{\sigma_{\text{strat}}}{\sigma_{\text{mkt}}} - 1.2\right)$$

$$\text{return\_penalty} = 1 + \frac{\text{gap}^2}{100}$$

$$\text{gap} = \max\big(0,\ (\mu_{\text{mkt}} - \mu_{\text{strat}}) \times 100 \times 252\big)$$

Key insight

The **quadratic** return penalty makes sustained underperformance expensive (falling 5% behind multiplies the denominator by $\sim$1.25). The strategy **defaults to index exposure** ($w = 1.0$) unless the signal is strong enough.

Data Description

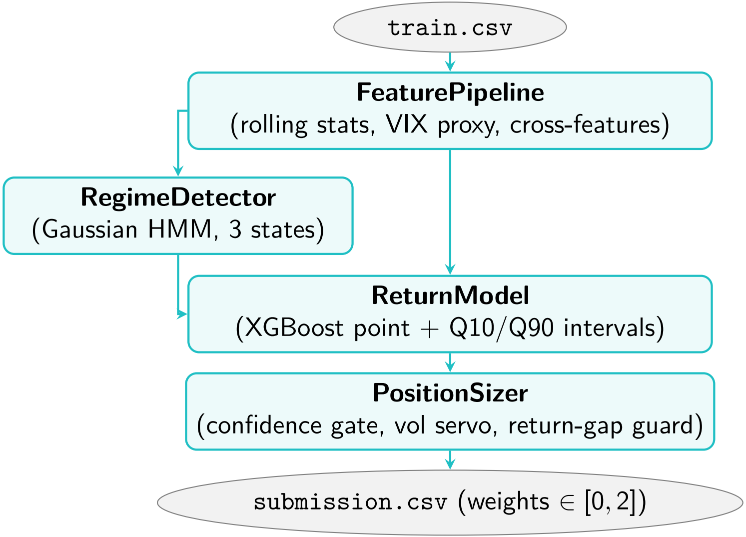
- **Train set**: 8 990 trading days with forward returns, risk-free rate, and anonymized features
  - Columns: I1–I5 (interest rates), E1–E12 (economic), S1–S5 (sentiment), P1–P11 (pricing), V1–V13 (volatility)
- **Project data**: Only `train.csv`, split chronologically (no random shuffling)

Temporal split (no random shuffling!)

Tuning block	Purge gap	Holdout
7 290 rows	200 rows	1 500 rows

The purge gap prevents autocorrelation leakage between train and evaluation data.

System Overview



Feature Engineering

Rolling Statistics

- Volatility: $\sigma_{5,20,60,252}$
- Momentum: $r_{5,20,60,252}$
- Sharpe ratios: $\mathrm{SR}_{20,60}$
- Drawdown: DD_{60}, MaxDD_{252}

VIX Features

- VIX Proxy: mean of V1–V13
- Percentile rank (train-frozen)
- Term-slope: V13 − V1
- VIX delta (history-buffered)

Cross-Features

- Yield-curve proxy: $I_{\text{last}} - I_{\text{first}}$
- Macro-sentiment divergence
- Price-momentum agreement
- Autocorrelation: $\rho_{5,20}$

Anti-Leakage

- Median imputation: train medians only
- 400-row history buffer for rolling windows
- Lagged targets only: `.shift(1)`

Total: ∼107 engineered features after availability filtering.

Volatility Proxy

- The V1–V13 columns in `train.csv` are anonymized volatility features
- A simple proxy maps V1–V13 into a unified volatility level
- Calculated as the cross-sectional mean: $\text{VIX Proxy}_t = \frac{1}{13}\sum_{i=1}^{13} V_{i,t}$

Why a proxy?

VIX is a well-known fear gauge. Compressing the volatility dimensions recovers a signal that is:

- Statistically associated with future volatility
- A useful volatility signal for this dataset
- Useful for regime detection and risk-aware position sizing

Four features are derived from the VIX proxy:

1. **VIX proxy**, cross-sectional mean level signal
2. **VIX percentile rank**, where current VIX sits vs. historical distribution
3. **VIX term-slope**, V13 − V1 (volatility term structure)

Market Regime Detection

- **Gaussian HMM** with 3 hidden states, diagonal covariance
- Inputs: `vix_proxy_pct`, `roll_vol_20`, `momentum_20`, `autocorr_20`

Output per row

- `regime_state`, Viterbi-decoded hidden state
- `regime_prob_{0,1,2}`, posterior probabilities
- `regime_stability`, confidence in regime assignment
- `regime_label`, semantic labels (e.g. `bull_low_vol`, `bear_high_vol`) assigned by training set mean return/volatility comparisons

Return Prediction Model (XGBoost)

Three models trained simultaneously on `market_forward_excess_returns`:

Model	Objective	Purpose
Point model	`reg:squarederror`	Direction & magnitude
Lower quantile (Q10)	`reg:quantileerror`, $\alpha = 0.1$	Lower bound
Upper quantile (Q90)	`reg:quantileerror`, $\alpha = 0.9$	Upper bound

Confidence signal:

$$\text{raw\_conf} = \frac{|\text{point\_pred}|}{\max(\text{Q90} - \text{Q10},\ \varepsilon)} \quad \longrightarrow \quad \text{confidence} = \text{clip}\left(\frac{\text{raw\_conf} - p_{10}}{p_{90} - p_{10}},\ 0,\ 1\right)$$

Percentiles p_{10}, p_{90} are computed on training data and frozen. This maps raw confidence to $[0, 1]$, filtering noise while preserving relative ordering.

Model Hyperparameters & Regularization

XGBoost parameters

- `max_depth = 4`
- `learning_rate = 0.05`
- `subsample = 0.8`
- `colsample_bytree = 0.8`

Tree Count & Regularization

- Point model uses **400 trees** to learn directional signal
- Quantile bounds are limited to **150 trees** ($\frac{1}{3}$ of point)
- Shorter ensembles act as early stopping to prevent overfitting to noisy tails

Purged Walk-Forward CV: Time-series cross-validation with a configurable purge gap between each training window and validation fold, preventing any forward information leakage through autocorrelated targets.

Confidence-Gated Position Sizing

Two-branch logic per time step:

$$\begin{cases} w = 1.0, & \text{if confidence} < \text{threshold} \\ w = 1.0 + \text{gain} \times \text{pred} \times \text{vol\_scale} \times \text{reg\_mult}, & \text{otherwise} \end{cases}$$

Volatility scaling (Moreira & Muir, 2017):

$$\text{vol\_scale} = \text{clip}\left(\frac{\sigma_{\text{target}}}{\sigma_t},\ 0.5,\ 2.0\right)$$

Regime multipliers

Regime	Multiplier
Bull, low volatility	1.30
Normal	1.00
Transition	0.70
Bear, high volatility	0.50

Rationale

The strategy stays at baseline exposure when confidence is low, scales risk inversely to volatility, and reduces active bets in unstable or high-volatility regimes.

Runtime Guards

Two sequential guards applied after the base weight calculation:

- **Volatility servo**: If rolling strategy vol / market vol exceeds 1.149 (soft cap) or 1.20 (hard cap), active deviation is scaled back to 55% or 20% to avoid the vol penalty.
- **Return-gap guard**: If strategy return falls behind the market, force $w \geq 1.0$ to prevent further divergence and avoid the quadratic return penalty.

Defensive Mechanism

The quadratic return penalty makes underperformance costly. Guards act defensively, reducing downside risk without removing the strategy's upside exposure.

Walk-Forward Backtesting

- Each CV fold re-fits the **entire modeling pipeline** independently
- Features, regime detector, and return model are all refit from scratch per fold
- No state, scaler, or model weight is shared between folds
- Model predictions are cached; strategy parameters are then scored with the exact competition metric

Fold-by-Fold Validation Diagnostics

Fold	Validation Range	Adj. Sharpe	Mean Weight	Dominant Regime
1	3 790 – 4 372	+0.9985	0.914	Transition
2	4 373 – 4 955	-0.2741	1.136	Transition
3	4 956 – 5 538	+0.8578	0.981	Bear (High Vol)
4	5 539 – 6 121	+1.5707	1.012	Transition
5	6 122 – 6 704	+0.9661	0.950	Bear (High Vol)
6	6 705 – 7 287	+0.9965	0.930	Transition

Strategy Tuning

- The model pipeline is fixed before tuning strategy parameters
- Inner CV fold predictions are cached once to avoid refitting the model for every trial
- Optuna runs **40 strategy-only trials** over confidence threshold, gain, target volatility, and soft volatility cap
- Initial untuned strategy mean CV adjusted Sharpe: **0.7246**
- Best inner CV mean adjusted Sharpe: **0.8526**

Why cache folds?

The expensive feature, regime, and return-model steps are fitted once per fold. Optuna then evaluates only the lightweight weight-generation rule, so the search remains reproducible and fast.

Holdout Results

Configuration	Adj. Sharpe	Ann. Return	Vol Ratio	Active Days
This model (Tuned)	**0.8605**	**+20.2%**	1.020	29.5%
Baseline ($w = 1.0$)	0.7151	+16.8%	1.000	–
Delta	**+0.1454**	**+3.4 pp**		

- **+20.3%** improvement in adjusted Sharpe over baseline on untouched holdout
- Inner CV mean adjusted Sharpe: **0.8526** (6 folds)
- Only 29.5% of days are active, so most days stay close to baseline exposure
- Tuned parameters: confidence threshold = 0.364, gain = 179.3, target vol = 0.298, soft vol cap = 1.149

Leakage Prevention

Risk	Mitigation
Feature leakage from future	`fit_transform` on training only; apply frozen stats
Rolling boundary leakage	400-row history buffer prepended to validation/holdout blocks
Temporal lookahead in CV	Purged walk-forward CV; separate 200-day purge before final holdout
Target leakage	Lagged targets only (`.shift(1)`); target excluded from features
Tuning lookahead	Hyperparameters tuned on inner temporal holdout
XGBoost non-determinism	Enforce `nthread=1`, `tree_method='hist'`, `seed=42`

Critical Importance

Strict anti-leakage controls prevent inflated backtest performance. Purge gaps and history buffering are mandatory for reliable time-series modeling.

Reproducibility & Design Notes

Reproducibility measures:

- All random seeds: 42 (XGBoost, sklearn, HMM)
- XGBoost: `nthread=1`, `tree_method='hist'` for cross-platform determinism
- Log `holdout_size` and `purge_gap` with every experiment

Why only 29.5% active days?

- The confidence threshold is high, as most days the prediction interval is wide relative to the point estimate
- Active days cluster around high-confidence signals: trend-consistent predictions during low-volatility bull regimes
- Index-tracking as the default absorbs return-gap risk; the strategy can only improve the score by being active

Summary

Key Takeaways:

- The adjusted Sharpe's quadratic return penalty fundamentally shapes the strategy design.
- Confidence-gating is essential: default to the index unless high confidence is present.
- Volatility management (Moreira & Muir, 2017) boosts Sharpe ratio by scaling risk dynamically.
- HMM-based regime detection provides context to optimize position sizing.
- Purged walk-forward validation is mandatory to estimate real out-of-sample performance.

References

References:

- Moreira, A. & Muir, T. (2017). *Volatility-Managed Portfolios*. Journal of Finance, 72(4), 1611–1644.
- Kaggle Competition: `https://www.kaggle.com/competitions/hull-tactical-market-prediction`
- Chen, T. & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD 2016.